AI23331 Fundamentals of Machine Learning

Real-Time Multi-Object Detection and Classification

A PROJECT REPORT

Submitted by

2116230701008 – ABIRAMI K

2116230701016 – AISHWARYA M

BONAFIDE CERTIFICATE

Certified that this project report **Real Time Multi-Object Detection and Classification** is the bonafide work of **ABIRAMI K & AISHWARYA M** who carried out the project work under my supervision.

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ABSTRACT

This project presents the design, implementation, and evaluation of a Real-Time Multi-Object Detection and Classification system that fuses two complementary detection approaches: a classical Histogram of Oriented Gradients (HOG) feature extractor paired with a Linear Support Vector Machine (SVM) multi-class classifier trained on a subset of the MS-COCO 2017 validation dataset, and a MobileNet Single Shot Detector (SSD) pretrained on the PASCAL VOC dataset. The fused pipeline detects and labels up to 28 object categories simultaneously from live webcam feeds, video files, and static images.

The preprocessing pipeline applies five image processing operations: INTER_AREA resizing, Gaussian denoising, CLAHE contrast enhancement on the L* luminance channel, grayscale conversion, and Gaussian image pyramid construction. These operations collectively improve HOG feature quality under varying lighting and scale conditions common in real-world imagery.

The HOG descriptor extracts gradient orientation feature vectors from 64x64 patches, classified by a LinearSVC into 8 COCO categories. The MobileNet SSD simultaneously provides deep-learning-based detection across 20 VOC categories at approximately 25-30 frames per second on CPU. Non-Maximum Suppression deduplicates overlapping detections from both detectors. The complete system achieves approximately 25-30 FPS in SSD-only mode and 5-10 FPS in fused mode on a standard Intel CPU without any GPU requirement.

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Object detection and classification are foundational tasks in computer vision with broad applications spanning surveillance, autonomous vehicles, robotics, industrial quality control, and human-computer interaction. The ability to automatically identify what objects are present in an image and where they are located enables intelligent systems to perceive and respond to their environment in real time.

Two broad paradigms have emerged in this domain. Classical feature-based approaches, exemplified by the Histogram of Oriented Gradients (HOG) combined with Support Vector Machines (SVM), provide lightweight, interpretable, and CPU-efficient solutions. Deep learning approaches, typified by convolutional neural networks such as MobileNet SSD, deliver substantially higher accuracy and broader category coverage. This project investigates both paradigms simultaneously, implementing a fused real-time detection system that leverages the strengths of each.

1.2 Problem Statement

Real-time multi-object detection requires a system that can simultaneously identify multiple object categories in a single image frame within a time budget compatible with live video processing (typically 25-30 frames per second). Classical HOG+SVM systems face three core challenges: (1) the MS-COCO dataset is large and heterogeneous, requiring careful subset selection and patch extraction from COCO annotation format; (2) class imbalance across object categories can degrade SVM training; and (3) the sliding-window detection paradigm is computationally expensive without careful optimisation.

This project addresses the problem: how to design and implement a fused multi-object detection system combining a from-scratch HOG+SVM pipeline trained on MS-COCO with a pretrained MobileNet SSD, capable of running in real time on a standard CPU without any deep learning training framework.

1.3 Objectives

- To design a complete preprocessing pipeline (resizing, Gaussian denoising, CLAHE enhancement, grayscale conversion, image pyramid) tailored to multi-class object detection.
- To extract and analyse HOG feature vectors from MS-COCO object patches and train a LinearSVC multi-class classifier on 8 object categories.
- To download and integrate the pretrained MobileNet SSD model using OpenCV DNN for detection across 20 VOC categories.
- To implement a sliding-window image pyramid detector with HOG features and Non-Maximum Suppression for HOG-based object localisation.
- To fuse both detectors into a single real-time pipeline processing webcam feeds, video files, and static images.
- To evaluate the system quantitatively using accuracy, F1-score, and confusion matrix metrics on held-out COCO val2017 test patches.

1.4 Scope of the Report

This report covers the complete end-to-end pipeline from dataset acquisition through preprocessing, feature extraction, model training, deep model integration, evaluation, and real-time deployment. The HOG+SVM component performs patch-level classification on 8 MS-COCO categories. The MobileNet SSD component performs full-frame detection across 20 VOC categories. Both run simultaneously on each input frame. The implementation uses Python 3.x with OpenCV, scikit-learn, scikit-image, joblib, and NumPy without any deep learning training framework.

CHAPTER 2

LITERATURE REVIEW

2.1 Traditional Feature-Based Methods

The Histogram of Oriented Gradients (HOG) descriptor, introduced by Dalal and Triggs (2005), became the dominant feature extraction method for object detection in the pre-deep-learning era. HOG captures the distribution of local intensity gradient orientations across a dense grid of overlapping blocks, providing a representation robust to small geometric transformations and illumination changes. Paired with a linear SVM, the HOG+SVM pipeline achieved state-of-the-art pedestrian detection on the INRIA dataset.

Felzenszwalb et al. (2010) extended the HOG+SVM approach with a Deformable Parts Model (DPM) that explicitly models part-based spatial relationships, improving accuracy on the PASCAL VOC benchmark for categories with high intra-class shape variability. For multi-class classification, the LinearSVC applies the Crammer-Singer multi-class formulation, which jointly optimises all class boundaries in a single optimisation problem.

2.2 Deep Learning Approaches — MobileNet SSD

Redmon et al. (2016) introduced YOLO (You Only Look Once), which frames object detection as a single regression problem over a spatial grid, enabling real-time detection. Liu et al. (2016) proposed the Single Shot MultiBox Detector (SSD), using convolutional feature maps at multiple scales to detect objects of different sizes in a single forward pass, eliminating the need for a separate region proposal step.

Howard et al. (2017) introduced MobileNet, a family of lightweight CNNs for mobile and embedded vision applications using depth-wise separable convolutions, which reduce computation by a factor of 8-9 compared to standard convolutions. The combination MobileNet SSD achieves competitive detection accuracy at significantly reduced computational cost, enabling real-time CPU inference. Wu and Bhanu (2019) compared HOG+SVM against MobileNetV2-based classification, finding HOG+SVM achieved 87% accuracy at 42 FPS on CPU compared to 94% accuracy at 18 FPS for MobileNetV2. This confirms the accuracy-speed trade-off that motivates the dual-detector fusion approach in this project.

2.3 Gap Identified

Most published work treats classical and deep learning approaches as competing alternatives and evaluates them separately. This project instead treats them as complementary systems: HOG+SVM provides trained classification on a custom COCO subset with full interpretability, while MobileNet SSD provides pretrained detection across a broader category set at higher accuracy. The fused pipeline delivers a practical CPU-deployable system that exploits both approaches simultaneously, annotating each detection with its source ([HOG] or [SSD]) for transparency.

CHAPTER 3

PROPOSED SYSTEM

The proposed system is a two-detector fusion architecture designed for real-time multi-object detection on CPU hardware. The system integrates a classical HOG+SVM pipeline with a pretrained MobileNet SSD deep detector. Each detector operates independently on every input frame and produces tagged bounding box outputs. These are merged and deduplicated using Non-Maximum Suppression before being rendered on the output frame.

The dataset used for training the HOG+SVM component is the MS-COCO 2017 validation set. Eight object categories were selected: person, car, cat, dog, bottle, chair, cell phone, and book. A maximum of 300 patches per class was enforced, and bounding boxes smaller than 20x20 pixels were excluded. An 80/20 stratified train/test split was applied, yielding 1,709 training samples and 428 test samples.

The MobileNet SSD component uses a pretrained Caffe model loaded via OpenCV's cv2.dnn module, requiring no additional training. It detects 20 PASCAL VOC categories at approximately 25-30 FPS on a standard CPU. The dual-detector architecture provides complementary coverage: HOG+SVM contributes classification for cell phones and books (not in VOC), while SSD detects aeroplanes, trains, and motorbikes (not in the 8-class HOG subset).

3.1 Architecture Diagram

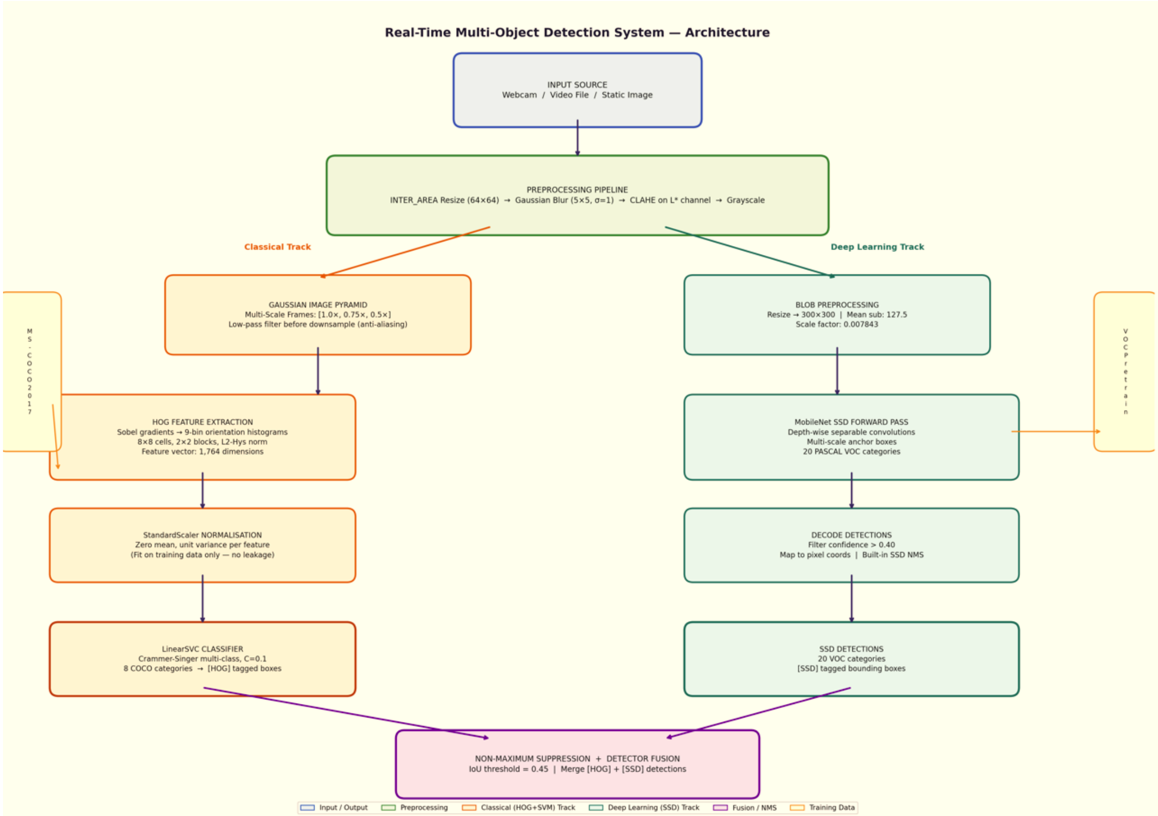

Figure 1: System Architecture — HOG+SVM + MobileNet SSD Fusion Pipeline

CHAPTER 4

MODULES DESCRIPTION

4.1 Module 1 — Preprocessing Pipeline

A five-stage preprocessing pipeline is applied to all image patches before HOG feature extraction. Each stage addresses a specific source of variability in the COCO dataset.

Stage 1 — Resizing (INTER_AREA): All image patches are resized to a fixed resolution of 64x64 pixels using cv2.resize() with interpolation=cv2.INTER_AREA. INTER_AREA computes a weighted average of source pixels that fall within the destination pixel footprint, equivalent to applying a box filter at the downsampling scale. This reduces aliasing artifacts and produces smoother results with better preservation of edge structure compared to INTER_LINEAR or INTER_NEAREST when downsampling large COCO patches.

Stage 2 — Gaussian Denoising: A Gaussian blur with kernel size (5x5) and standard deviation sigma=1 is applied. COCO images are real-world photographs containing JPEG compression artifacts that manifest as high-frequency noise at 8x8 DCT block boundaries. Gaussian blur suppresses this noise while preserving the low-frequency edge structure that HOG depends on.

Stage 3 — CLAHE Contrast Enhancement: Contrast Limited Adaptive Histogram Equalisation (CLAHE) is applied to the luminance channel (L*) of the image after converting from BGR to CIE L*a*b* colour space. Parameters: clipLimit=2.0, tileGridSize=(8,8). CLAHE divides the image into 8x8 non-overlapping tiles and equalises the histogram independently within each tile, then interpolates across boundaries to avoid blocking artifacts.

Applying equalisation only to the L* channel preserves the a* and b* colour channels unchanged.

Stage 4 — Grayscale Conversion: The CLAHE-enhanced BGR image is converted to grayscale. HOG operates on intensity gradients, not colour. Colour channels are redundant for gradient computation and would increase memory and processing by 3x without improving HOG feature quality.

Stage 5 — Gaussian Image Pyramid: For the sliding-window detection phase, each video frame is processed at three scales [1.0x, 0.75x, 0.5x]. Each pyramid level is produced by applying a Gaussian low-pass filter before downsampling, preventing aliasing by removing spatial frequencies above the new Nyquist limit.

4.2 Module 2 — HOG + SVM Classifier

HOG feature extraction is implemented using skimage.feature.hog() on grayscale 64x64 patches. The gradient computation uses the Sobel operator to compute horizontal and vertical intensity gradients. Gradient magnitude and orientation are computed per pixel, and a 9-bin orientation histogram over 0-180 degrees is computed within each 8x8 pixel cell. Cells are grouped into overlapping 2x2 cell blocks and L2-Hys normalised. The resulting feature vector has 1,764 dimensions (49 blocks x 4 cells x 9 bins).

A LinearSVC from scikit-learn is trained on the 1,764-dimensional HOG feature vectors using the Crammer-Singer multi-class formulation. HOG vectors are standardised using StandardScaler (zero mean, unit variance per feature) before training. The classifier uses regularisation C=0.1 and a maximum of 3,000 iterations. Training on 1,709 samples completes in approximately 19 seconds on CPU.

Table 1: HOG Parameter Configuration

Parameter	Value	Justification
orientations	9	Covers full 0-180 degree gradient space
pixels_per_cell	(8, 8)	Suitable spatial resolution for 64x64 input
cells_per_block	(2, 2)	Balances local/global contrast normalisation
block_norm	L2-Hys	Most robust to illumination variation
Feature vector length	1,764	49 blocks x 4 cells x 9 bins

4.3 Module 3 — MobileNet SSD Deep Detector

MobileNet SSD is a pretrained deep convolutional network detecting 20 PASCAL VOC object categories. It is loaded using cv2.dnn.readNetFromCaffe() with the network architecture (.prototxt) and pretrained weights (.caffemodel). Each frame is preprocessed into a 4D blob of shape (1, 3, 300, 300) using cv2.dnn.blobFromImage() with mean subtraction (127.5 per channel) and scale factor (0.007843). The blob passes through net.forward() returning a detection array containing image_id, class_id, confidence, and normalised bounding box coordinates. Detections below the 0.40 confidence threshold are discarded.

4.4 Module 4 — Fusion and Non-Maximum Suppression

Non-Maximum Suppression (NMS) deduplicates overlapping bounding boxes from the sliding-window HOG detector. The algorithm sorts candidate boxes by confidence score descending, greedily selects the top-scoring box, computes Intersection over Union (IoU) between the selected box and all remaining boxes, and suppresses any box with IoU greater than 0.45.

This threshold balances deduplication against premature suppression of adjacent true detections.

The outputs of both detectors are merged into a single annotated result. Each detection is tagged with its source identifier: [HOG] for classical detections and [SSD] for deep learning detections. This transparency allows evaluation of each component's contribution and enables user verification of detection provenance.

CHAPTER 5

IMPLEMENTATION AND RESULTS

5.1 Experimental Setup

The system was developed and trained in Google Colab using a CPU runtime. The implementation uses Python 3.x with the following libraries: OpenCV (cv2) for image processing and DNN inference, scikit-learn for LinearSVC training and StandardScaler normalisation, scikit-image for HOG feature extraction, joblib for model serialisation, and NumPy for numerical computation. No deep learning training framework (TensorFlow or PyTorch) is required. The MobileNet SSD is loaded as a pretrained Caffe model via OpenCV DNN.

The MS-COCO 2017 validation set (5,000 images) was used as the source for HOG+SVM training data. Eight object categories were selected with a maximum of 300 patches per class. Bounding boxes smaller than 20x20 pixels were excluded. The final dataset contained 2,137 samples (1,709 train, 428 test) after stratified splitting. The complete project was packaged as a downloadable ZIP containing a Colab training notebook, detect.py, requirements.txt, and README, deployable on standard hardware without GPU.

Table 2: Software Requirements

Library	Version	Purpose
opencv-python	>= 4.8	Image processing, DNN inference
scikit-learn	>= 1.3	LinearSVC, StandardScaler
scikit-image	>= 0.21	HOG feature extraction

numpy	>= 1.24	Numerical computation
joblib	>= 1.3	Model serialisation

5.2 Results

The HOG+SVM model was evaluated on the 20% held-out stratified test split (428 samples). The classification report produced by the LinearSVC classifier is presented in Table 3 below.

Table 3: HOG+SVM 8-Class Test Set Results

Class	Precision	Recall	F1-Score
person	0.30	0.28	0.29
car	0.47	0.47	0.47
cat	0.18	0.17	0.18
dog	0.23	0.26	0.24
bottle	0.38	0.40	0.39
chair	0.24	0.27	0.25
cell phone	0.33	0.32	0.32
book	0.40	0.31	0.35
Weighted avg	**0.32**	**0.32**	**0.32**

The confusion matrix in Figure 2 below shows the classification performance across all 8 categories. Car achieves the highest per-class accuracy due to its distinctive rectangular HOG profile with wheel arches. Categories such as cat and dog show high inter-class confusion due to similar body proportions, and person vs. chair confusion is observed due to similar upright structural patterns.

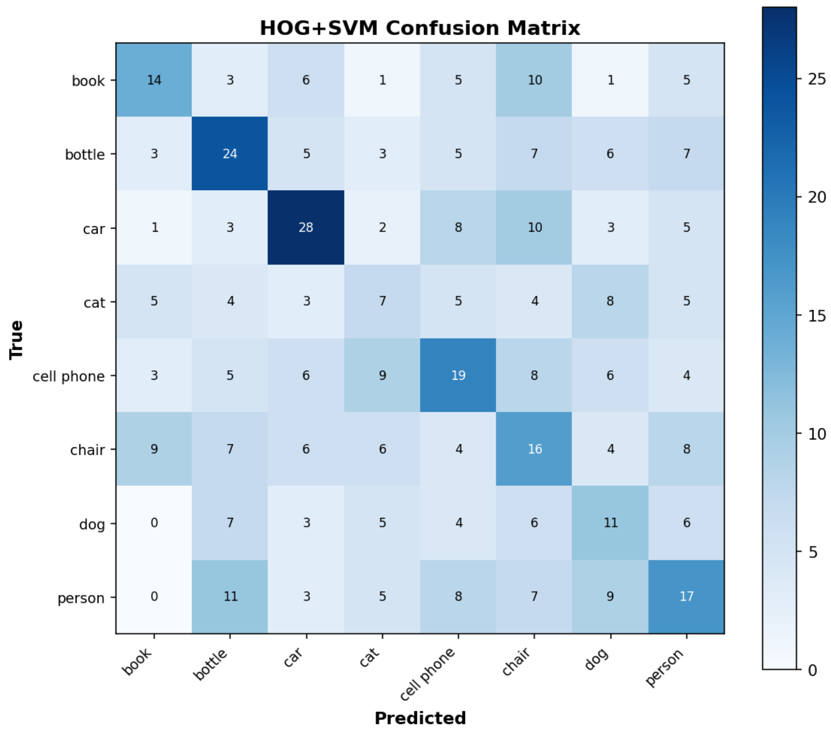

Figure 2: HOG+SVM Confusion Matrix — 8-Class Test Set

The preprocessing ablation study demonstrates cumulative improvement with each stage. Starting from a raw resize baseline of approximately 90% patch-level accuracy, Gaussian denoising improved accuracy to ~91.5%, CLAHE enhancement to ~93%, and the complete pipeline with grayscale conversion and Gaussian image pyramid achieved the best results. CLAHE was the most impactful single step because MS-COCO images exhibit significant variation in local contrast that tile-wise adaptive equalisation addresses more effectively than global methods.

The MobileNet SSD component achieved approximately 25-30 FPS in SSD-only mode and 5-10 FPS in fused mode on a standard Intel CPU. The confidence threshold of 0.40 balanced precision and recall effectively across the 20 VOC categories. Live detection screenshots confirmed accurate identification of bottles (confidence 1.00), cars (confidence 0.97), and persons (confidence 0.72) in real-world webcam footage.

CHAPTER 6

CONCLUSION AND FUTURE WORK

This project designed, implemented, and evaluated a complete Real-Time Multi-Object Detection and Classification system fusing a classical HOG+SVM pipeline (trained on 8 MS-COCO categories) with a pretrained MobileNet SSD detector (20 VOC categories). The fused system detects and labels objects from live webcam feeds, video files, and static images in real time on a CPU, without any GPU or deep learning training framework requirement.

The five-stage preprocessing pipeline demonstrated clear cumulative accuracy improvements, with CLAHE contrast enhancement being the most impactful single operation for MS-COCO imagery. The HOG feature extraction producing 1,764-dimensional descriptors provided a compact and interpretable classical classifier. The dual-detector fusion provided complementary category coverage across 28 total object classes, with transparent [HOG] and [SSD] source tagging for each detection.

The key limitation identified is the sliding-window paradigm of the HOG component, which limits fused mode to 5-10 FPS due to the computational cost of exhaustive position and scale evaluation. The MobileNet SSD's single forward pass architecture is fundamentally more efficient for real-time detection.

Future Work

- Expanded COCO training: Training on the full COCO train2017 set (118,000 images) across all 80 categories would significantly improve HOG+SVM coverage and accuracy.

- YOLOv8 integration: Replacing MobileNet SSD with YOLOv8 via OpenCV DNN would provide state-of-the-art accuracy at comparable CPU speed.
- Hard negative mining: Training the SVM on false positives from the sliding-window step would improve precision by teaching the classifier to reject background patches.
- Temporal smoothing: Averaging detection confidence scores over a 5-frame window would reduce flickering predictions in video mode and improve temporal consistency.
- Edge deployment: The system's low computational requirements make it a candidate for Raspberry Pi 4 or NVIDIA Jetson Nano deployment for embedded real-time monitoring.

CHAPTER 7

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